

OmniXRI TinyML 小學堂 2025



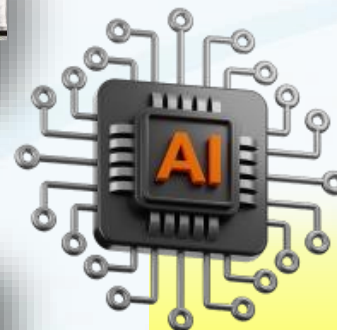
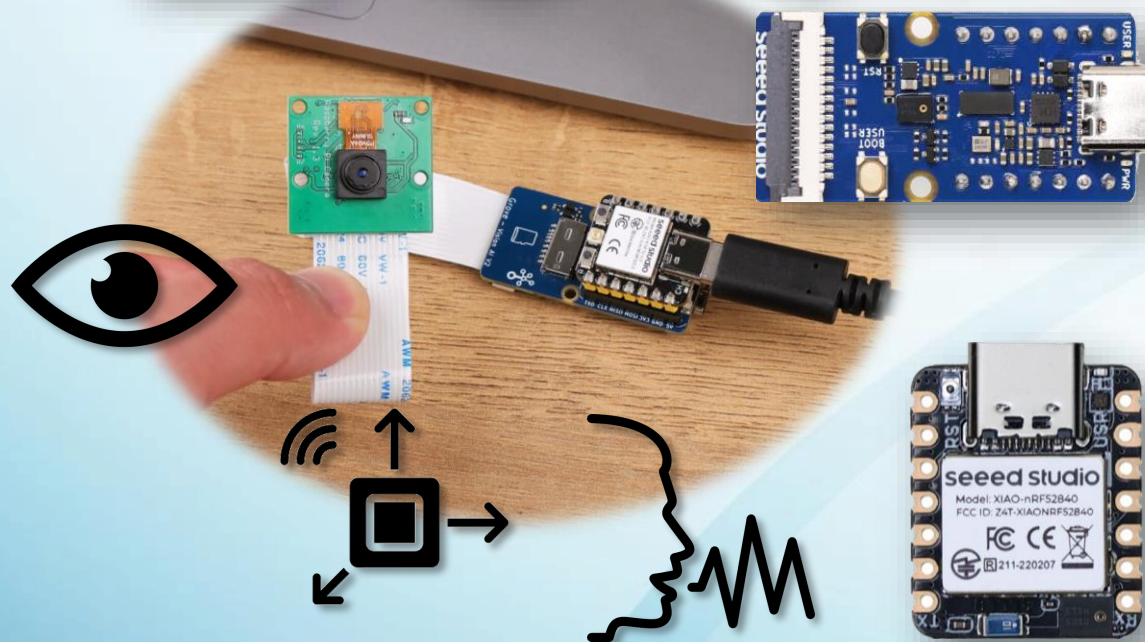
歡迎加入
邊緣人俱樂部



沒有最邊



只有更邊



Cortex-M
Processor

Ethos-U
MicroNPU

【第 11 講】

影像應用—物件偵測



歐尼克斯實境互動工作室 (OmniXRI Studio)
許哲豪 (Jack Hsu)

簡報大綱



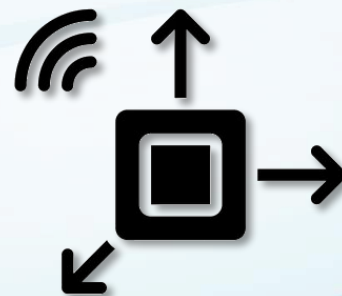
- 11.1. Edge Impulse 開發環建置
- 11.2. 影像資料集建置
- 11.3. 模型選用與訓練
- 11.4. 模型部署與測試

本課程完全免費，請勿移作商業用途！

歡迎留言、訂閱、點讚、轉發，讓更多需要的朋友也能一起學習。

完整課程大綱：<https://omnixri.blogspot.com/2025/03/omnixri-tinymml-2025-0.html>

課程直播清單：<https://www.youtube.com/@omnixri1784/streams>



11.1. Edge Impulse 開發環建置

TinyML 開發流程選項

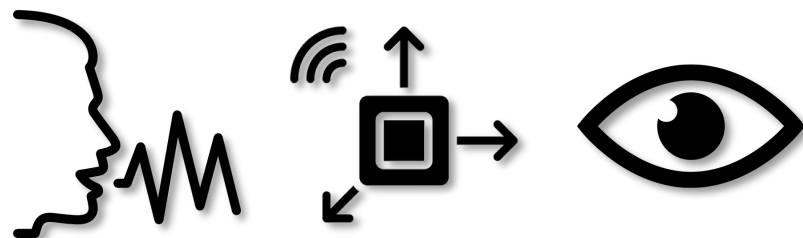
模型訓練工具名稱

適用感測器

主要特色



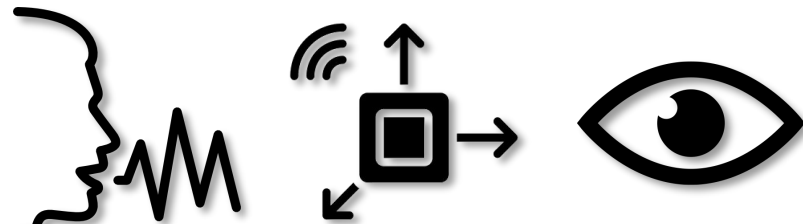
參考【第 8 講】聲音辨識應用—環境音辨識



免費版：資料集需自行建置，僅提供模型訓練及轉成 TensorFlow Lite for Microcontroller 格式。可利用雲端 Google Colab 或本地端 Jupyter Notebooks 進行模型訓練及部署。



參考【第 6 講】TinyML 開發流程



免費版：可支援各式感測器及開發板，提供一定額度資料集建置、資料前處理（特徵提取）模型訓練、模型優化及多種部署方式。亦可導入已訓練好的 TensorFlow Lite, ONNX 等模型。

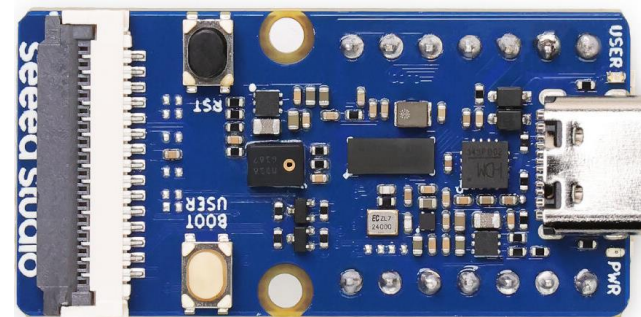
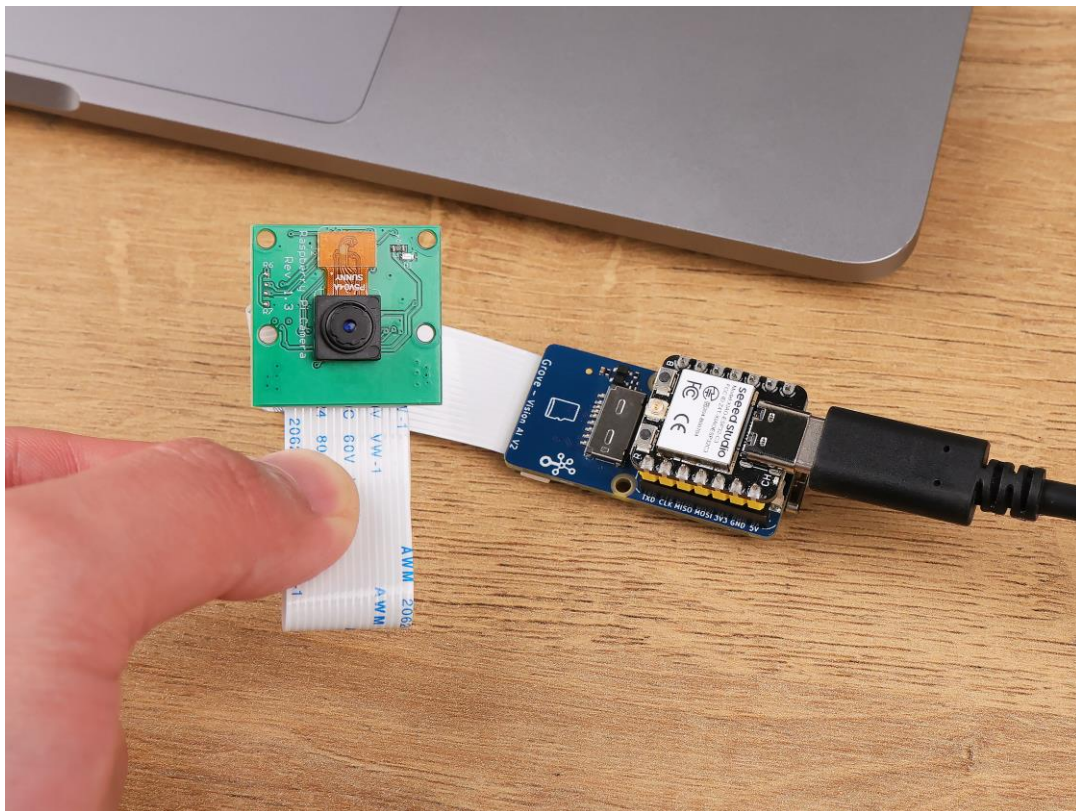


參考【第 10 講】影像應用—影像分類

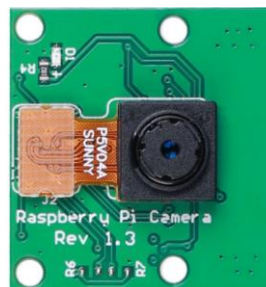


免費版：影像分類可多分類、物件偵測同一模型僅能一類。事件觸發只能設一種條件。可支援雲端資料收集、模型訓練及部署，亦導入外部已訓練好之 TensorFlow Lite 模型。

Seeed Grove Vision AI V2 Kit



**Grove Vision AI
Module V2**
(Himax WiseEye2
HX6538)



**OV5647-62
Camera Module
for Pi3B+4B**
(含軟排)

2529 x 1944 (5MP)

**Xiao ESP32C3
(WIFI / BT5)**
(無線通訊用，選配)



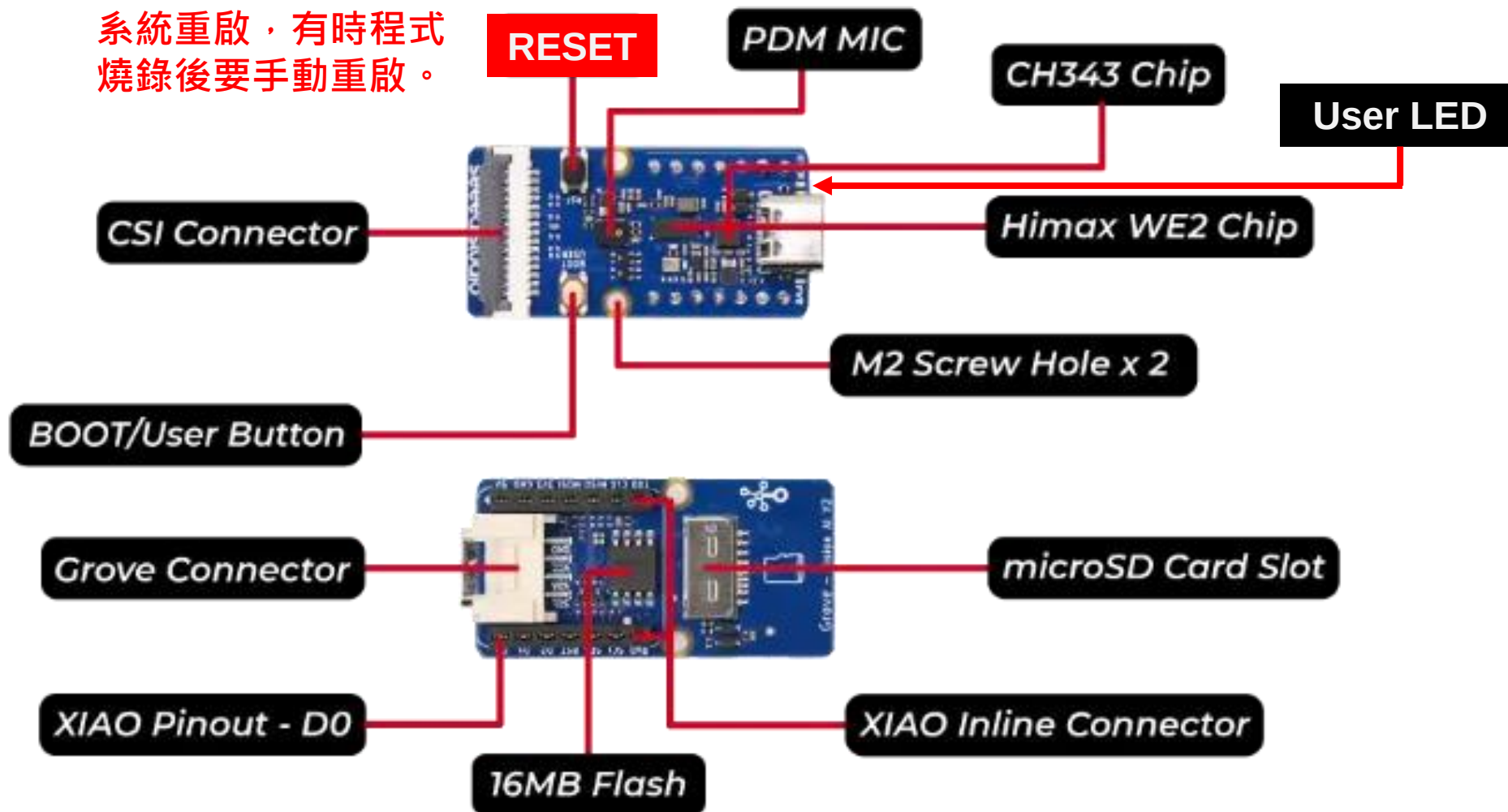
資料來源：https://wiki.seeedstudio.com/grove_vision_ai_v2/

Grove Vision AI Module V2 技術文件



https://wiki.seesstudio.com/grove_vision_ai_v2/

Grove Vision AI Module V2 元件圖



資料來源：https://wiki.seeedstudio.com/grove_vision_ai_v2/

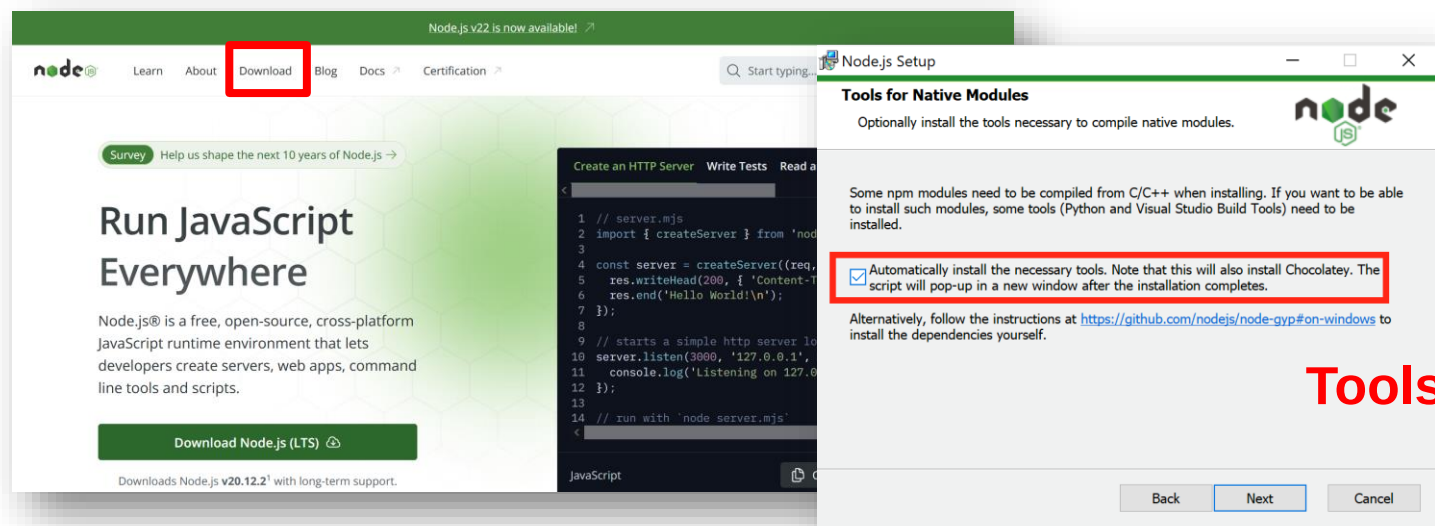
下載及安裝必要工具

1. Python 3.x



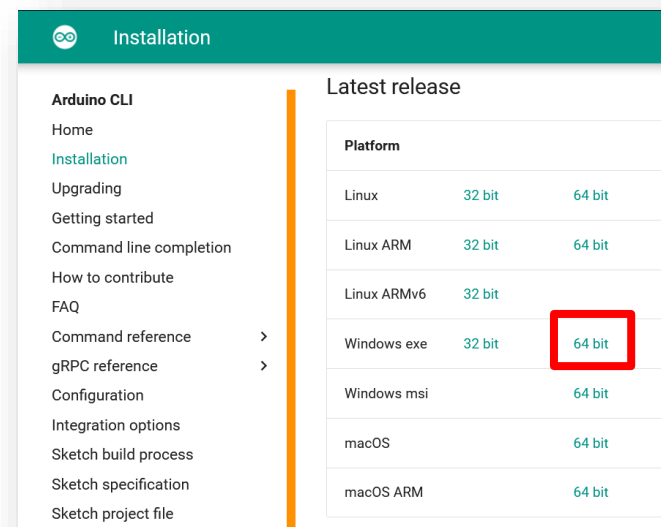
2. Node.js

(若後續 Edge Impulse 無法安裝
可換回舊版，例：v18.20.2 LTS)



3. Arduino CLI

(選配，依不同開發板需求)



安裝時要注意
記得一定要勾選

Tools for Native Modules

以 `node -v` 及 `npm -v`
檢查安裝好的版本

安裝 Edge Impulse 工作環境

- 安裝Edge Impulse CLI
`npm install -g edge-impulse-cli --force`
- 安裝完成後會得到下列工具
 - **edge-impulse-daemon** 透過 COM 埠和雲端平台連接
 - `edge-impulse-uploader` 允許上傳本機檔案
 - `edge-impulse-data-forwarder` 透過 COM 埠上傳開發板感測器數值
 - `edge-impulse-run-impulse` 顯示裝置上運行的模型
 - `edge-impulse-blocks` 建立自定義區塊
 - **himax-flash-tool** 燒錄 Haimax WE1 & WE2 開發板

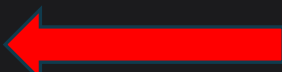
<https://docs.edgeimpulse.com/docs/tools/edge-impulse-cli/cli-installation>

安裝驅動程式 (USB Virtual COM)

Driver

If you find that the Grove Vision AI V2 is not recognised after connecting it to your computer. Then you may need to install the CH343 driver on your computer. Here are some links to download and install the CH343 driver.

- Windows Vendor VCP Driver One-Click Installer: CH343SER.EXE
- Windows Vendor VCP Driver: CH343SER.ZIP
- Windows CDC driver one-click installer: CH343CDC.EXE
- Windows CDC driver: CH343CDC.ZIP
- macOS Vendor VCP Driver: CH34xSER_MAC.ZIP

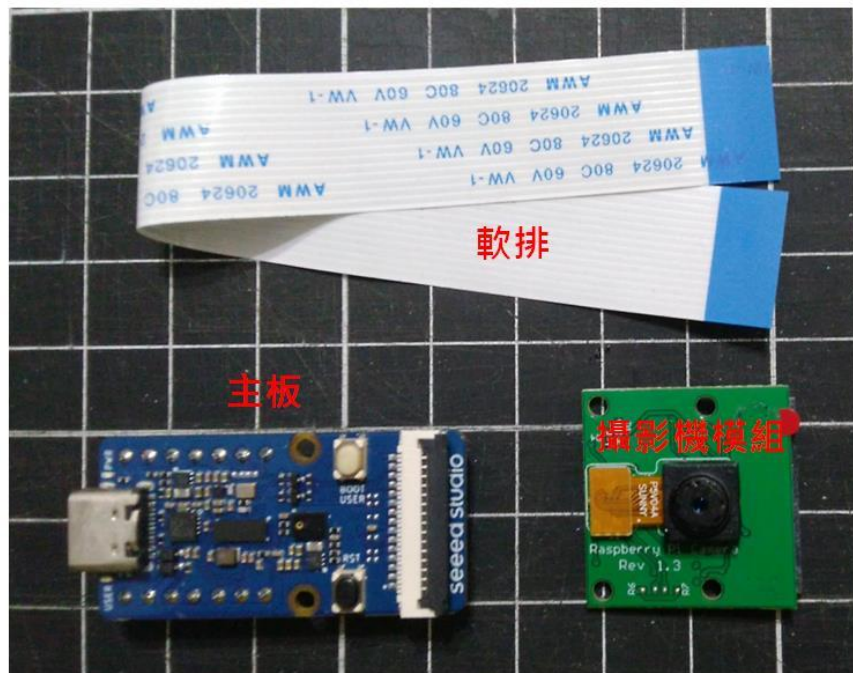


只需安裝一次
之後可略過

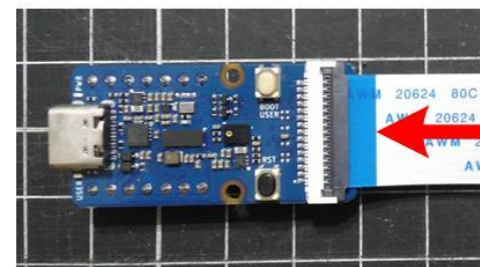
Windows環境 https://wiki.seeedstudio.com/grove_vision_ai_v2/

開發板組裝 — 連接攝影機

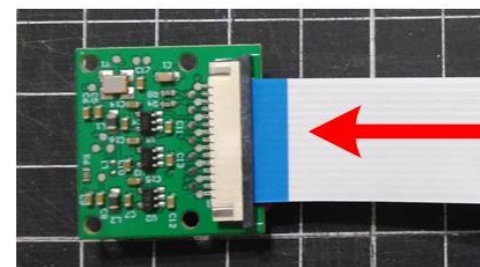
Seeed Grove Vision AI Module V2 & Camera



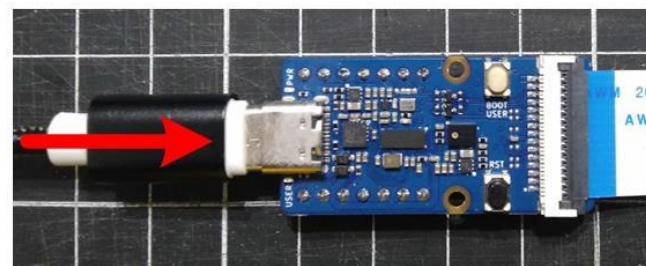
註：插入軟排時請注意正反面



1. 軟排插入
主板



2. 軟排插入
攝影機模組

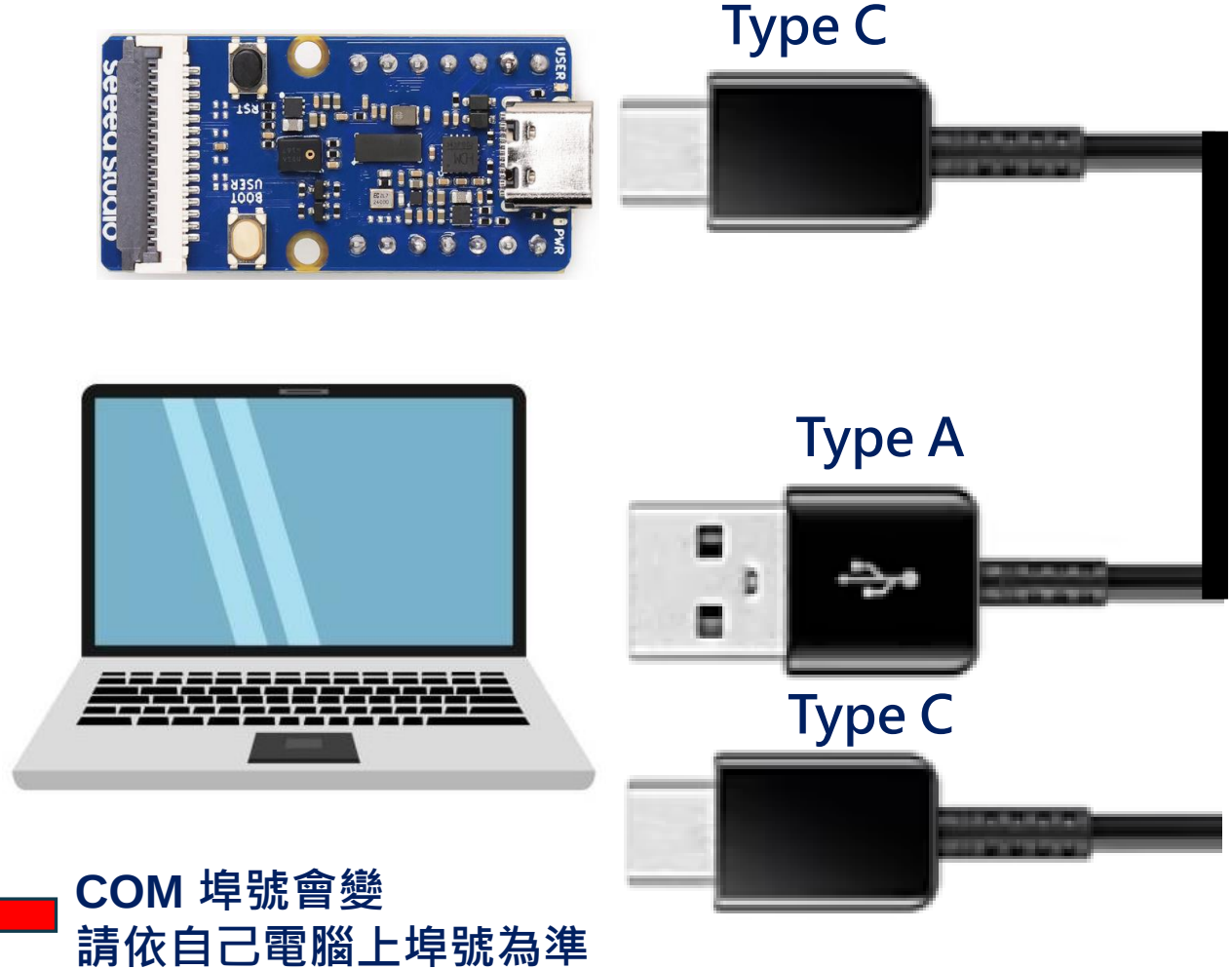


3. USB插入
主板

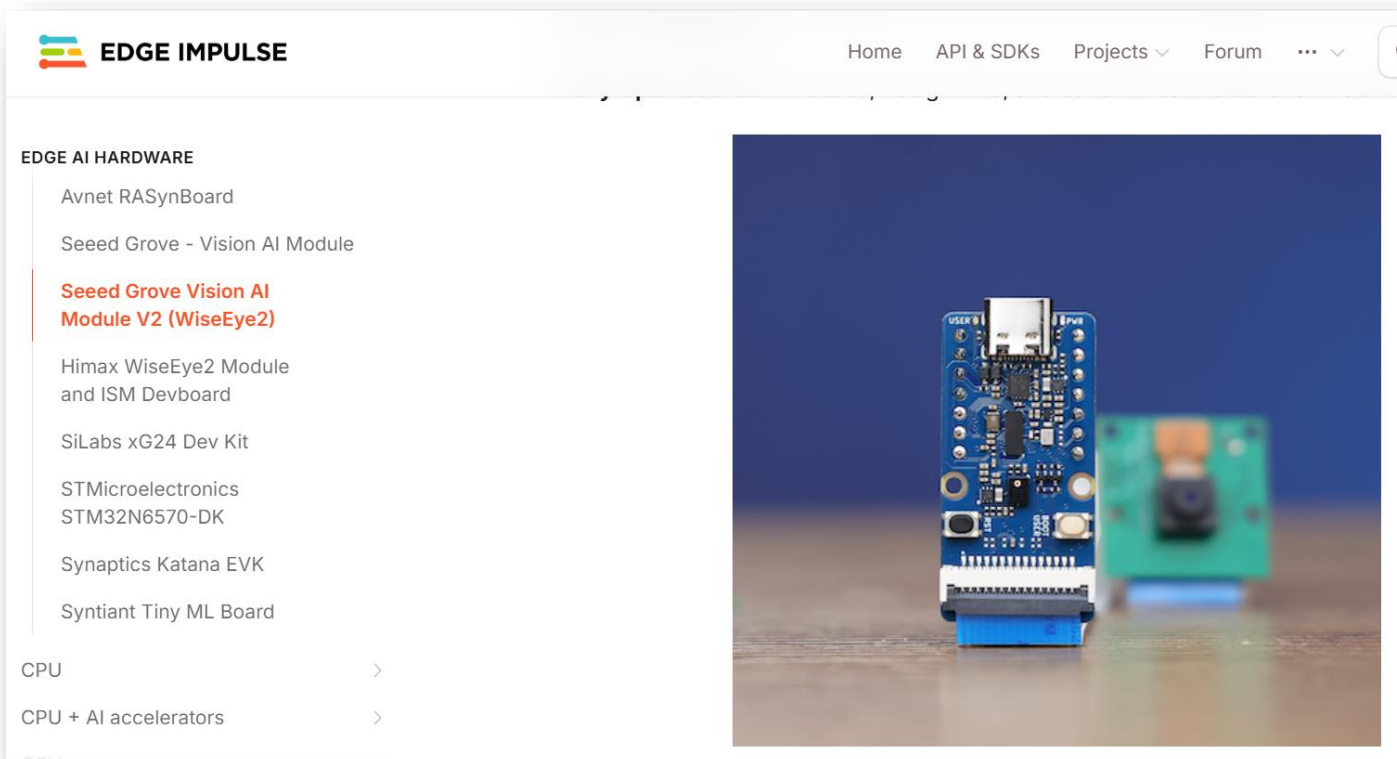
OmniXRI整理製作, 2024/07/15

資料來源：<https://omnixri.blogspot.com/2024/07/vmaker-edge-ai-19-mcunpu.html>

開發板組裝 — 連接USB



Edge Impulse AI 硬體 — WiseEye2說明



下載 **Edge Impulse**
預編譯映像檔及工具，
方便將開發板連接到
雲端進行資料擷取。

<https://cdn.edgeimpulse.com/firmware/seeed-grove-vision-ai-module-v2.zip>

檢查 CLI 版本 (至少 **1.27.1**)
`edge-impulse-daemon --version`

<https://docs.edgeimpulse.com/docs/edge-ai-hardware/mcu-+-ai-accelerators/himax-seeed-grove-vision-ai-module-v2-wise-eye-2>

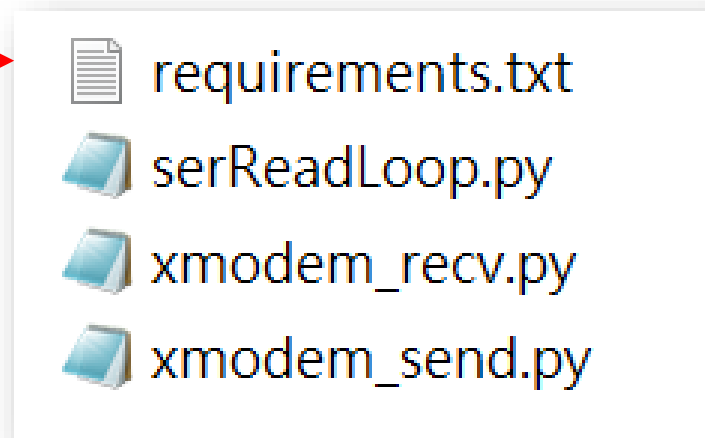
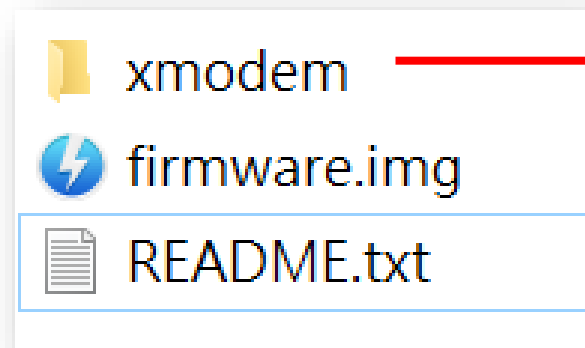
安裝 COM 高速通訊工具 xmodem

seeed-grove-vision-ai-module-v2.zip 解壓縮後

COM 高速通訊

預編譯映像檔

燒錄說明文件



缺少 `flash_windows.bat` 且說明文件不夠正確。

進入 Windows 命令列模式，執行下列命令安裝 xmodem 及必要套件。
`pip install -r xmodem/requirements.txt`

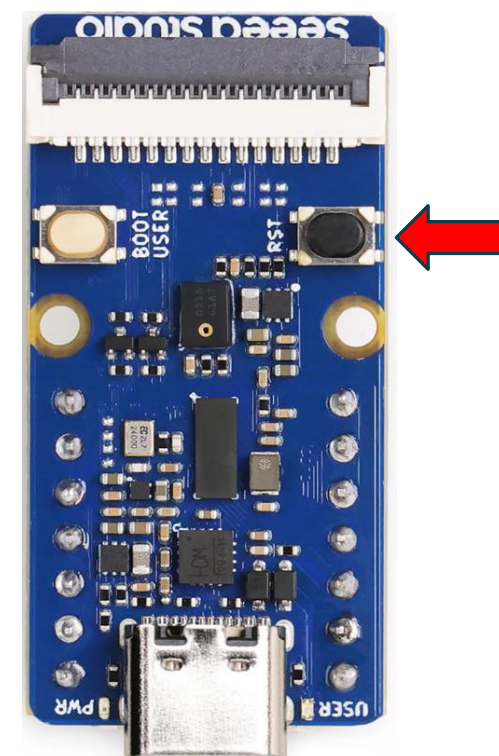
燒錄預編譯映像檔 — Edge Impulse CLI

使用 Edge Impulse CLI 工具燒錄

himax-flash-tool -d WiseEye2 -f firmware.img

```
C:\Users\jack_\Downloads\seed-grove-vision-ai-module-v2 (1)>himax-flash-tool -d WiseEye2 -f firmware.img
[HMX] Connecting to COM6...
[HMX] Connected, press the **RESET** button on your Seeed Grove Vision AI Module V2 now
[HMX] Restarted into bootloader. Sending file.
[HMX] Sending 3488 blocks
100% | ETA: 0s | 3488/3488
[HMX] Sent all blocks (NAK count: 0)
[HMX] Press **RESET** to start the application...
[HMX] Could not verify application Failed to get a valid response (looking for "Compiler Version") from device. Response was: "
1st BL Modem Build DATE=Nov 30 2023, 0x0002000b
Please input any key to enter X-Modem mode in 100 ms
waiting input key...
slot flash_offset 0x00100000
New MemDesp himax_sec_SB_image_process PASS
set_memory_s_ns
bl_status = 0x800000, HX_DSP_FLAG 1
bl_status = 0x800000
jump_addr=0x10000000
1st BL Modem Build DATE=Nov 30 2023, 0x0002000b
Please input any key to enter X-Modem mode in 100 ms
waiting input key...
slot flash_offset 0x00100000
New MemDesp himax_sec_SB_image_process PASS
set_memory_s_ns
bl_status = 0x800000, HX_DSP_FLAG 1
bl_status = 0x800000
jump_addr=0x10000000
"
[HMX] Did you press RESET on the board?
```

**等待 Grove Vision
AI V2 按下 Reset**

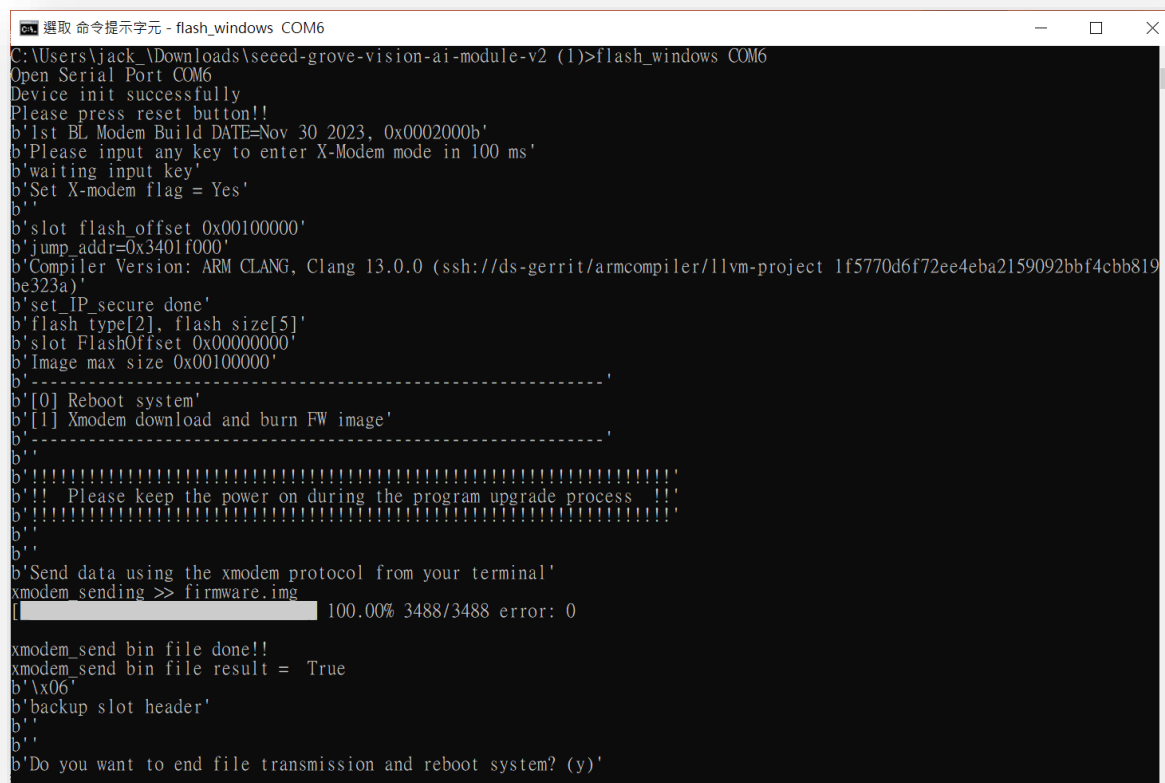


燒錄預編譯映像檔 — Python + xmodem

進入 Windows 命令列模式，執行下列 Python 程式。

`python xmodem/xmodem_send.py --port=COM6 --baudrate=921600 --protocol=xmodem --file=firmware.img`

請依實際 COM 埠號輸入

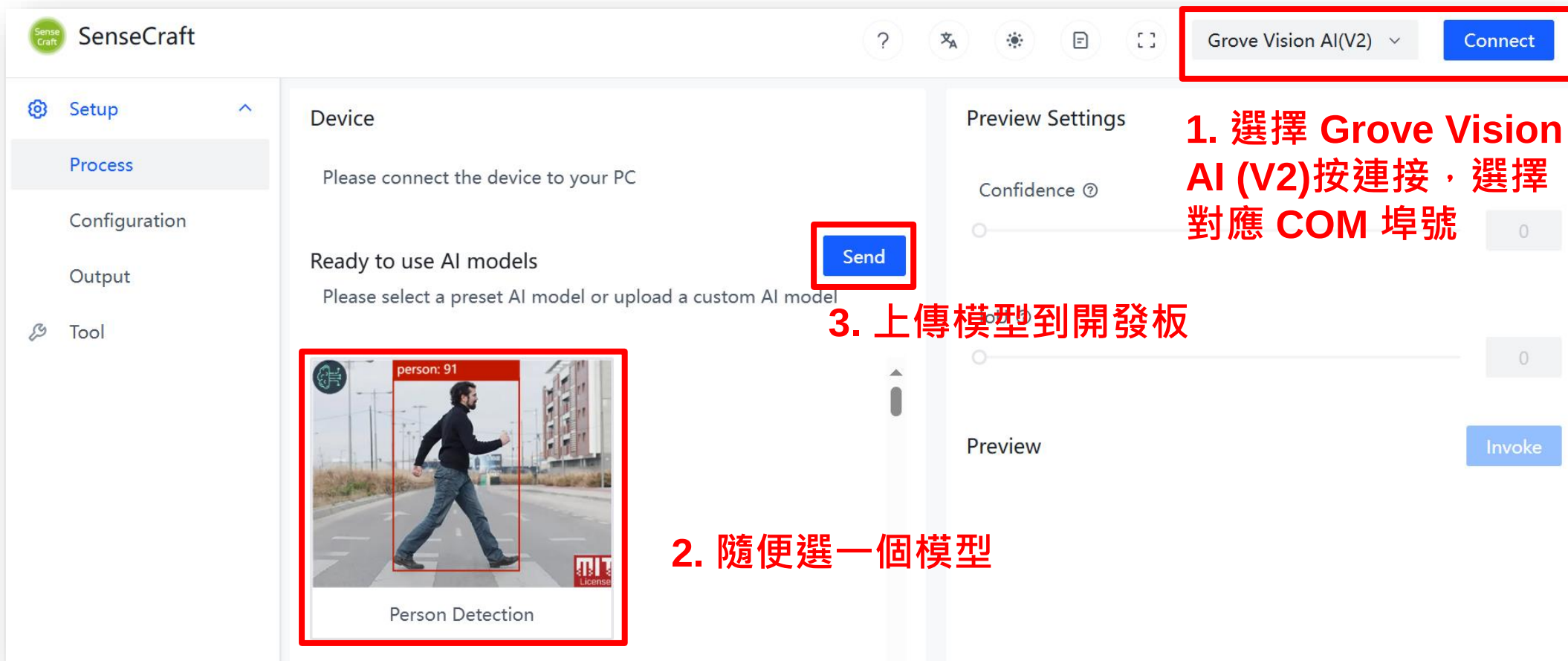


以上命令可以建立 **flash_windows.bat**，方便下次使用。

執行後出現如左側畫面，待詢問是否重啟時，按下開發板 **RESET** 鍵，再關閉命令視窗即完成燒錄。

燒錄方式二選一即可。

回復出廠設定 — SenseCraft Web Toolkit



The screenshot shows the SenseCraft Web Toolkit interface. On the left is a sidebar with navigation links: Setup, Process (highlighted), Configuration, Output, and Tool. The main area is divided into three sections: Device, Ready to use AI models, and Preview Settings. The Device section has a 'Send' button highlighted with a red box. The Ready to use AI models section shows a 'Person Detection' model with a red box around the image and the text 'person: 91'. The Preview Settings section has a 'Grove Vision AI(V2)' dropdown and a 'Connect' button highlighted with a red box. Red text annotations are present: '1. 選擇 Grove Vision AI (V2)按連接，選擇對應 COM 埠號' (1. Choose Grove Vision AI (V2) click connect, choose corresponding COM port number) pointing to the dropdown and 'Connect' button; '2. 隨便選一個模型' (2. Choose any model) pointing to the 'Person Detection' model; and '3. 上傳模型到開發板' (3. Upload model to development board) pointing to the 'Send' button.

SenseCraft

Setup

Process

Configuration

Output

Tool

Device

Please connect the device to your PC

Ready to use AI models

Please select a preset AI model or upload a custom AI model

Send

Preview Settings

Confidence

0

0

Preview

Invoke

1. 選擇 Grove Vision AI (V2)按連接，選擇對應 COM 埠號

2. 隨便選一個模型

3. 上傳模型到開發板

<https://seeed-studio.github.io/SenseCraft-Web-Toolkit/>

開發板連接 Edge Impulse Studio

使用 Edge Impulse CLI 工具連接雲端專案

edge-impulse-daemon --clean (不加 --clean 可以延用上一次資料)

```
C:\Users\jack_\Downloads\seed-grove-vision-ai-module-v2 (1)>edge-impulse-daemon --clean
Edge Impulse serial daemon v1.32.1
? What is your user name or e-mail address (edgeimpulse.com)? omnixri@gmail.com
? What is your password? [hidden]
Endpoints:
  Websocket: wss://remote-mgmt.edgeimpulse.com
  API: https://studio.edgeimpulse.com
  Ingestion: https://ingestion.edgeimpulse.com

[SER] Connecting to COM6
[SER] Serial is connected, trying to read config...
[SER] Clearing configuration
[SER] Clearing configuration OK
[SER] Retrieved configuration
[SER] Device is running AT command version 1.8.0

? To which project do you want to connect this device? (↵ type to search) (Press <enter> to submit)
OmniXRI / Tutorial: Recognize sounds from audio-1
OmniXRI / Seed Grove Vision AI V2 - PDM
OmniXRI / Visual Anomaly Grove Vision AI Module V2
> OmniXRI / Tutorial: object detection
OmniXRI / Coscup 2023 Gesture Remote Controller
OmniXRI / Xiao_nRF52840_IMU_Test
OmniXRI / Tutorial: Recognize sounds from audio
(Move up and down to reveal more choices)

? To which project do you want to connect this device? (↵ type to search) 701733
Setting upload host in device... OK
Configuring remote management settings... OK
Configuring API key in device... OK
Configuring HMAC key in device... OK
[SER] Device is not connected to remote management API, will use daemon
[WS] Connecting to wss://remote-mgmt.edgeimpulse.com
[WS] Connected to wss://remote-mgmt.edgeimpulse.com
? What name do you want to give this device? Grove Vision AI V2
[WS] Device "Grove Vision AI V2" is now connected to project "Tutorial: object detection". To connect to another project, run `edge-impulse-daemon --clean`.
[WS] Go to https://studio.edgeimpulse.com/studio/701733/acquisition/training to build your machine learning model!
```

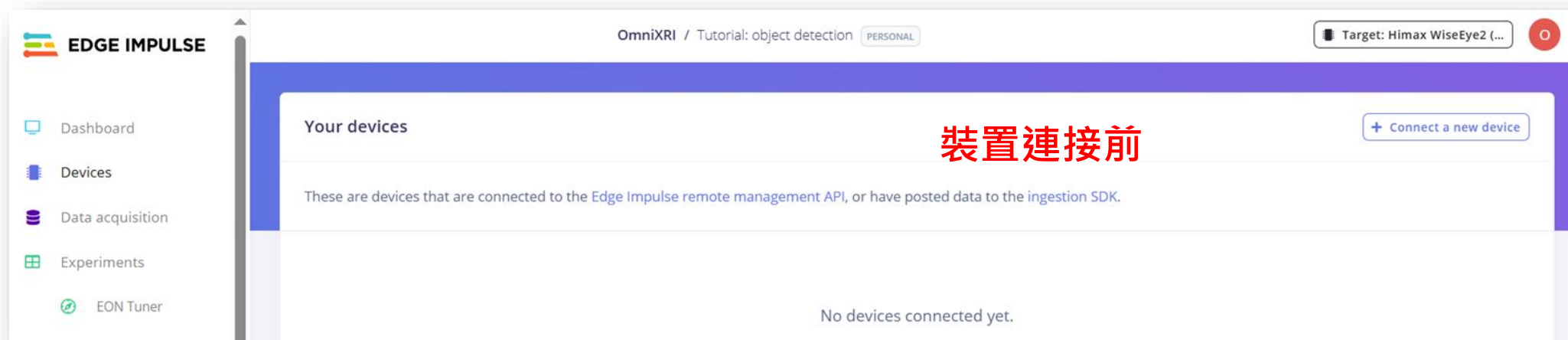
輸入帳號密碼

上下移動選擇專案

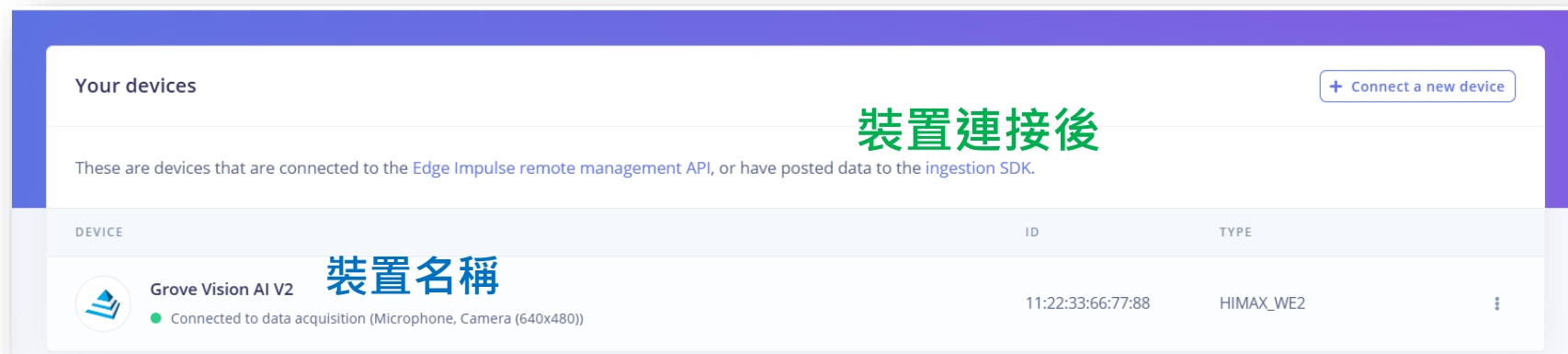
裝置只能連接
已存在之專案

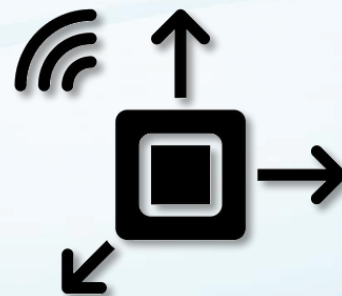
輸入裝置名稱

Edge Impulse Studio 裝置清單



綠燈表連線中
紅燈表未連線





11.2. 影像資料集建置

影像資料集比較

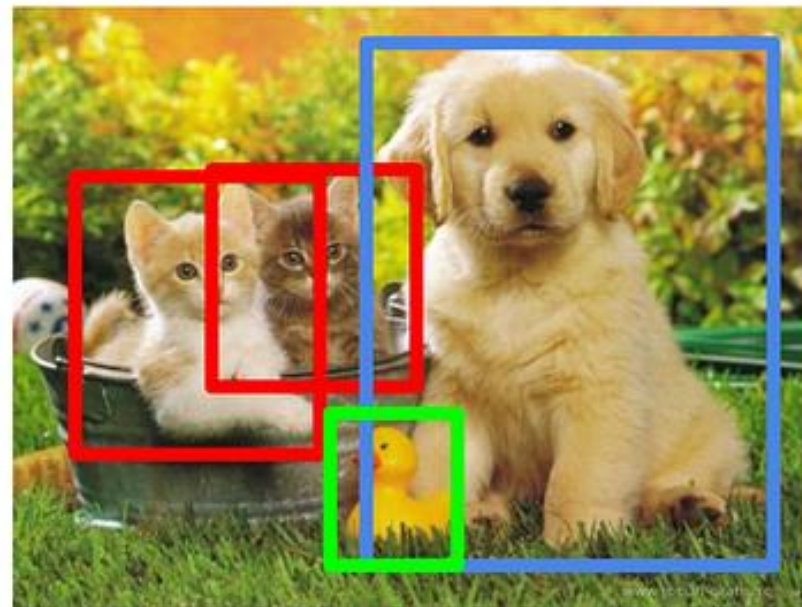
影像分類 (Image Classification)



CAT

一張影像只有一個主體物體，
整張影像只分成一類（標籤）。

物件偵測 (Object Detection)



CAT, DOG, DUCK

一張影像可以有多個物體，
分別框出不同物件並給不同標籤。

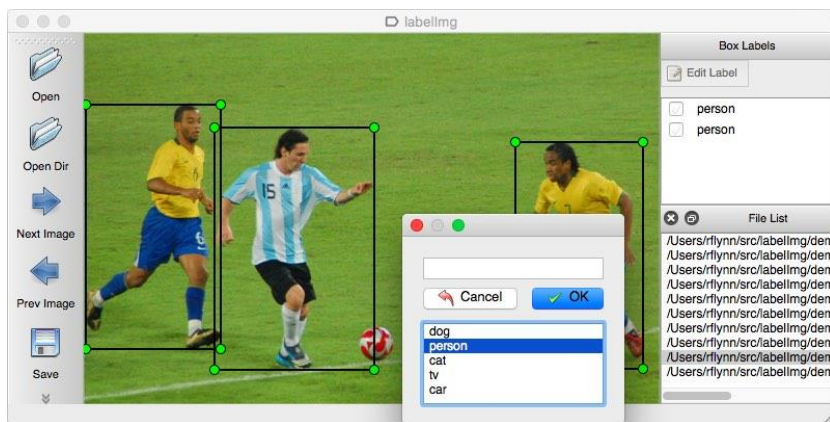
常見物件偵測資料標註工具



<https://github.com/wkentaro/labelme>



<https://www.cvat.ai/>



<https://github.com/HumanSignal/labelimg>



<https://roboflow.com/>

常見物件偵測標註格式 — VOC, YOLO

VOC 物件偵測資料格式

```

<size> 影像尺寸
  <width>353</width> 影像寬度
  <height>500</height> 影像高度
  <depth>3</depth> 色彩深度(通道數)
</size>
<segmented>0</segmented> 是否分割
<object> 物件 (目標) 訊息
  <name>dog</name> 物件名稱 (20分類)
  <pose>Left</pose> 拍攝角度 (前後左右及未定)
  <truncated>1</truncated> 是否被遮擋或截斷
  <difficult>0</difficult> 是否難以檢測
  <bndbox> 物件 (目標) 外框
    <xmin>48</xmin> 物件左上角X座標
    <ymin>240</ymin> 物件左上角Y座標
    <xmax>195</xmax> 物件右下角X座標
    <ymax>371</ymax> 物件右下角Y座標
  </bndbox>
</object>
  
```

邊界框位置
以左上表示

YOLO 物件偵測資料轉換

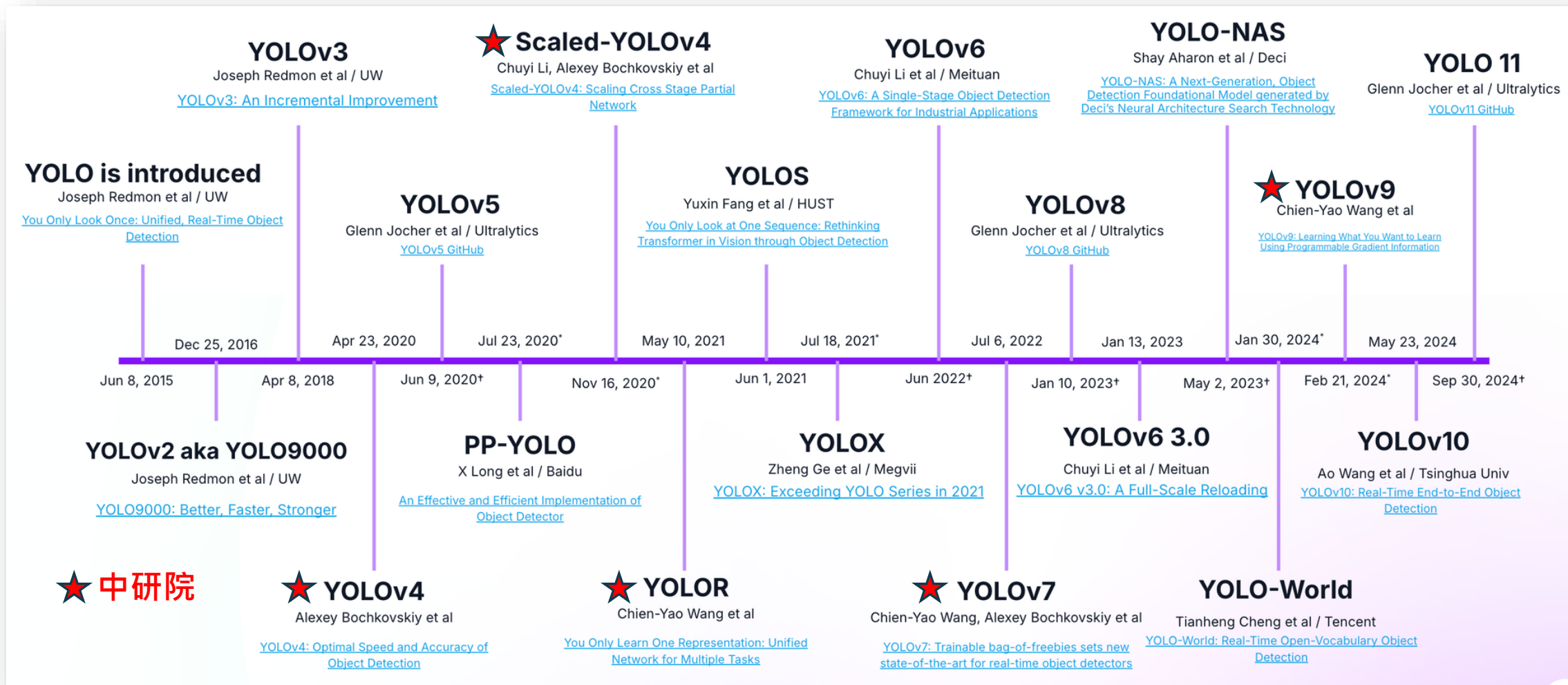
$$\begin{aligned}
 X &= (x_{min} + (x_{max} - x_{min}) / 2) * 1.0 / width \\
 Y &= (y_{min} + (y_{max} - y_{min}) / 2) * 1.0 / height \\
 W &= (x_{max} - x_{min}) * 1.0 / width \\
 H &= (y_{max} - y_{min}) * 1.0 / height
 \end{aligned}$$

邊界框位置
以中心表示

YOLO 物件標註檔案格式(*.txt)

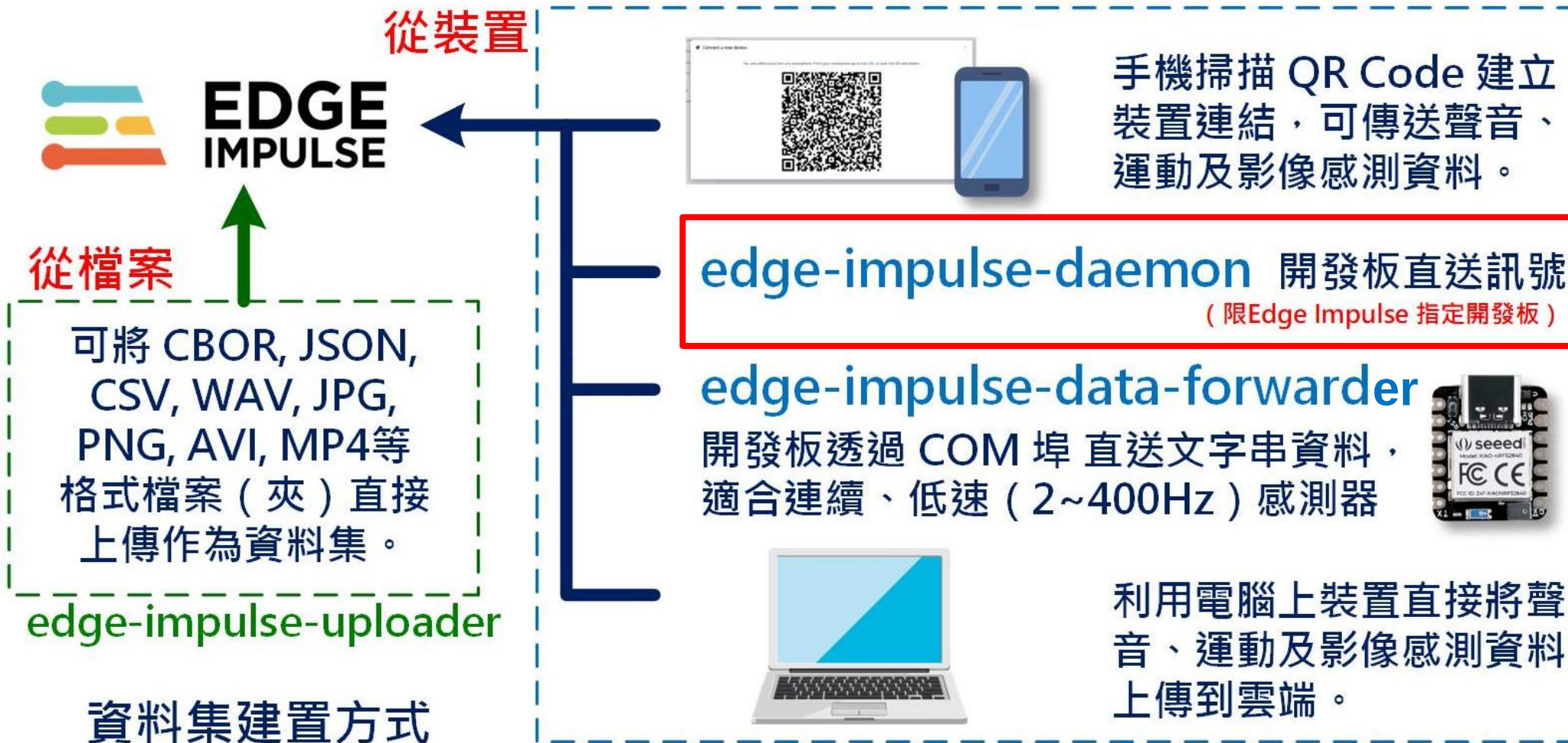
[分類編號ID] [物件中心X座標] [物件中心Y座標] [物件寬度佔比W] [物件高度佔比H]
 0 0.344192634561 0.611000000000 0.416430594901 0.262000000000

YOLO 家族發展史



資料來源：<https://blog.roboflow.com/guide-to-yolo-models/>

Edge Impulse 資料集建置方式



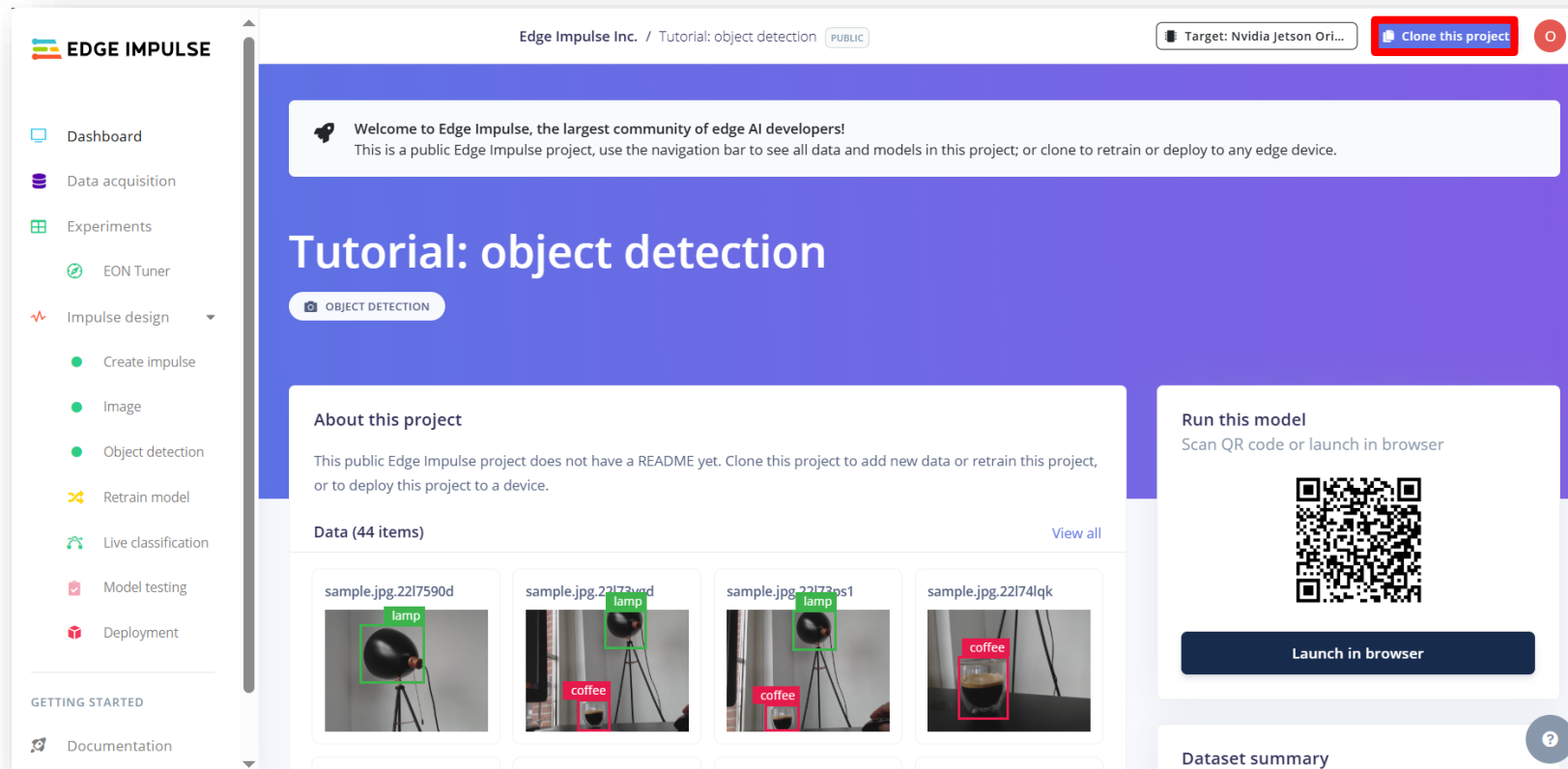
資料集建置方式

OmniXRI整理製作, 2025/04/16

Edge Impulse 物件偵測範例

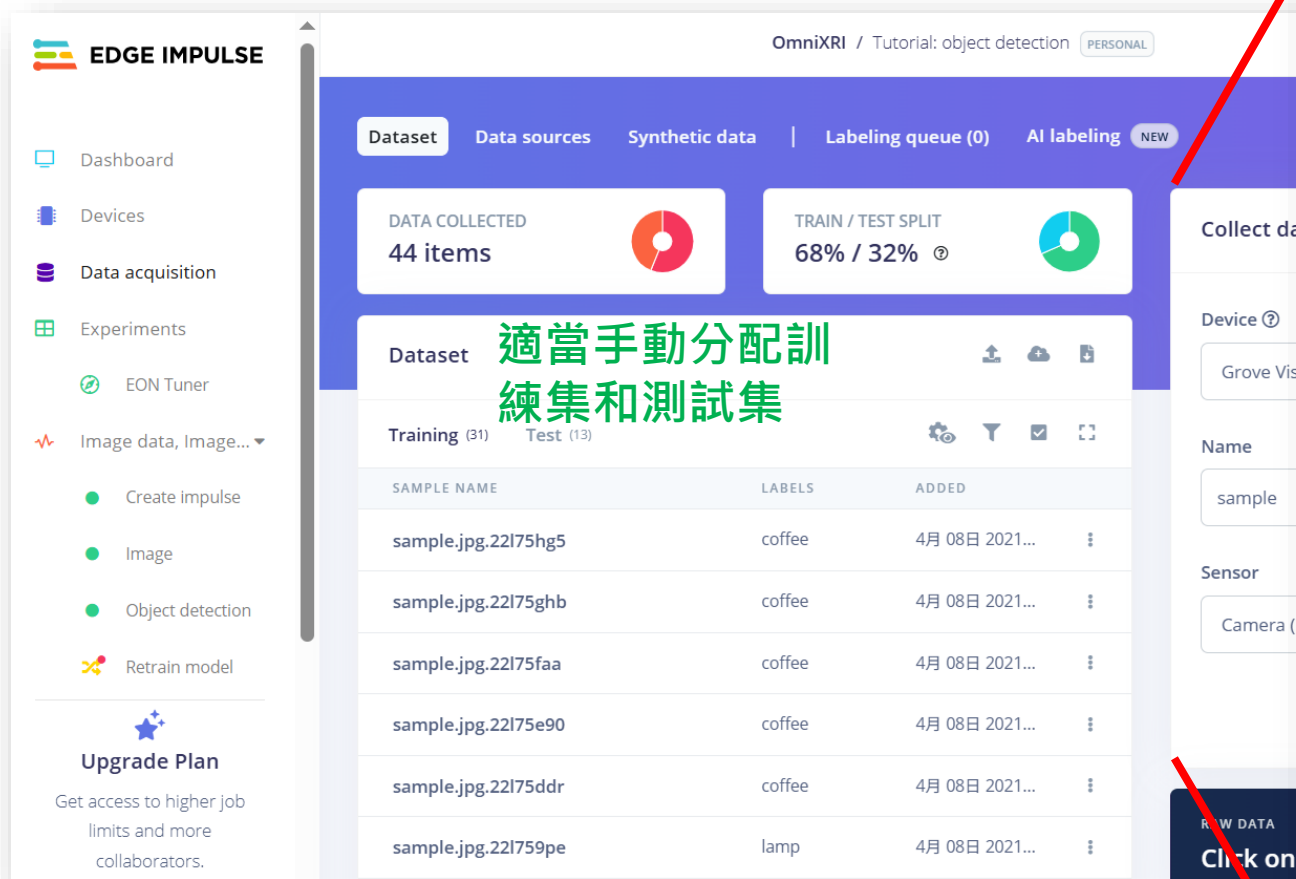
Tutorial: object detection (咖啡、燈泡)

複製專案到個人專案區



<https://docs.edgeimpulse.com/docs/tutorials/end-to-end-tutorials/computer-vision/object-detection/object-detection>

Edge Impulse Studio 資料擷取



EDGE IMPULSE

OmniXRI / Tutorial: object detection PERSONAL

Dataset Data sources Synthetic data Labeling queue (0) AI labeling NEW

DATA COLLECTED 44 items

TRAIN / TEST SPLIT 68% / 32%

Dataset 適當手動分配訓練集和測試集

SAMPLE NAME	LABELS	ADDED
sample.jpg.22l75hg5	coffee	4月 08日 2021...
sample.jpg.22l75ghb	coffee	4月 08日 2021...
sample.jpg.22l75faa	coffee	4月 08日 2021...
sample.jpg.22l75e90	coffee	4月 08日 2021...
sample.jpg.22l75ddr	coffee	4月 08日 2021...
sample.jpg.22l759pe	lamp	4月 08日 2021...

Training (31) Test (13)

Upgrade Plan

Get access to higher job limits and more collaborators.

Collect data 若無裝置連接則此區域空白，無法操作。

Device ?

Grove Vision AI V2 裝置名稱

Label

test 標籤名稱

Sensor

Camera (640x480) 選擇攝影機

Microphone

Camera (640x480)

Camera feed

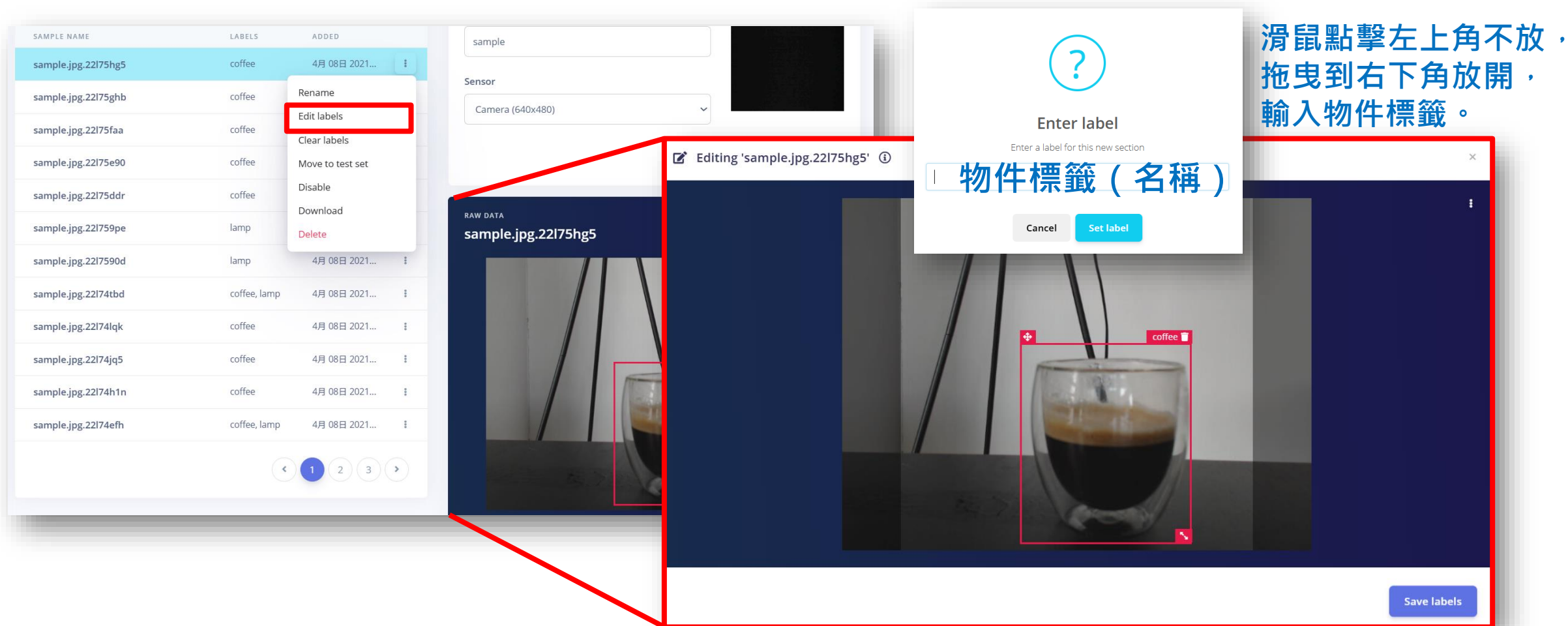


開始取像

Start sampling

強烈建議要以裝置上攝影機取像，不要以網路上取得高畫質影像當資料集。

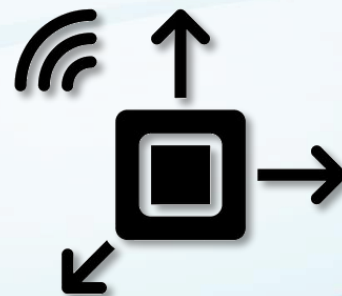
Edge Impulse Studio 建立物件標籤



The screenshot displays the Edge Impulse Studio interface. On the left, a table lists sample images and their labels. A context menu is open for the first sample, with 'Edit labels' highlighted. The main area shows a camera feed of a coffee cup. A red box highlights the 'Edit labels' button in the bottom right. A modal window titled 'Enter label' is open, showing the text '物件標籤 (名稱)' and a 'Set label' button. A red arrow points from the text '滑鼠點擊左上角不放，拖曳到右下角放開，輸入物件標籤。' to the top-left corner of the modal window.

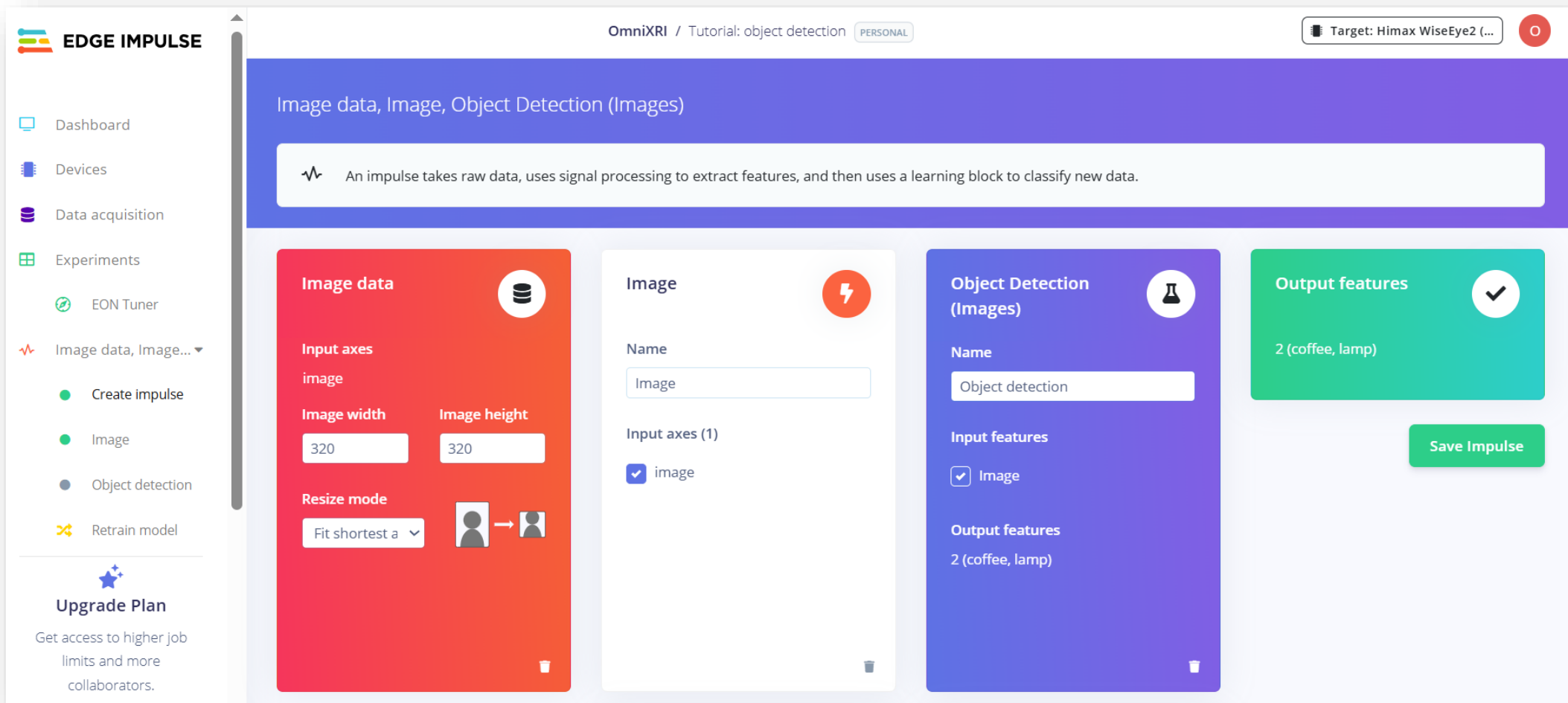
SAMPLE NAME	LABELS	ADDED
sample.jpg.22l75hg5	coffee	4月 08日 2021...
sample.jpg.22l75ghb	coffee	
sample.jpg.22l75faa	coffee	
sample.jpg.22l75e90	coffee	
sample.jpg.22l75ddr	coffee	
sample.jpg.22l759pe	lamp	
sample.jpg.22l7590d	lamp	4月 08日 2021...
sample.jpg.22l74tbd	coffee, lamp	4月 08日 2021...
sample.jpg.22l74lqk	coffee	4月 08日 2021...
sample.jpg.22l74jq5	coffee	4月 08日 2021...
sample.jpg.22l74h1n	coffee	4月 08日 2021...
sample.jpg.22l74efh	coffee, lamp	4月 08日 2021...

滑鼠點擊左上角不放，拖曳到右下角放開，輸入物件標籤。



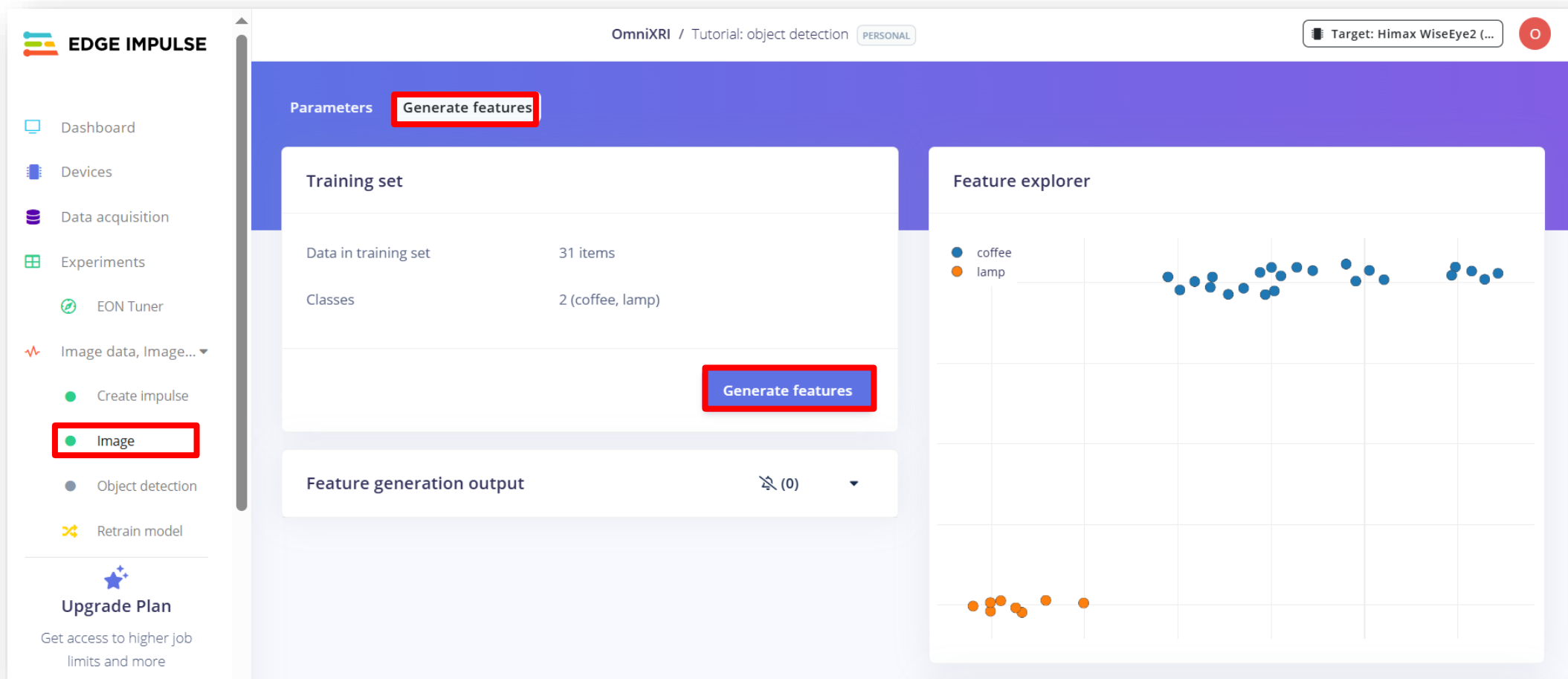
11.3. 模型選用與訓練

Edge Impulse Studio — 建立模型



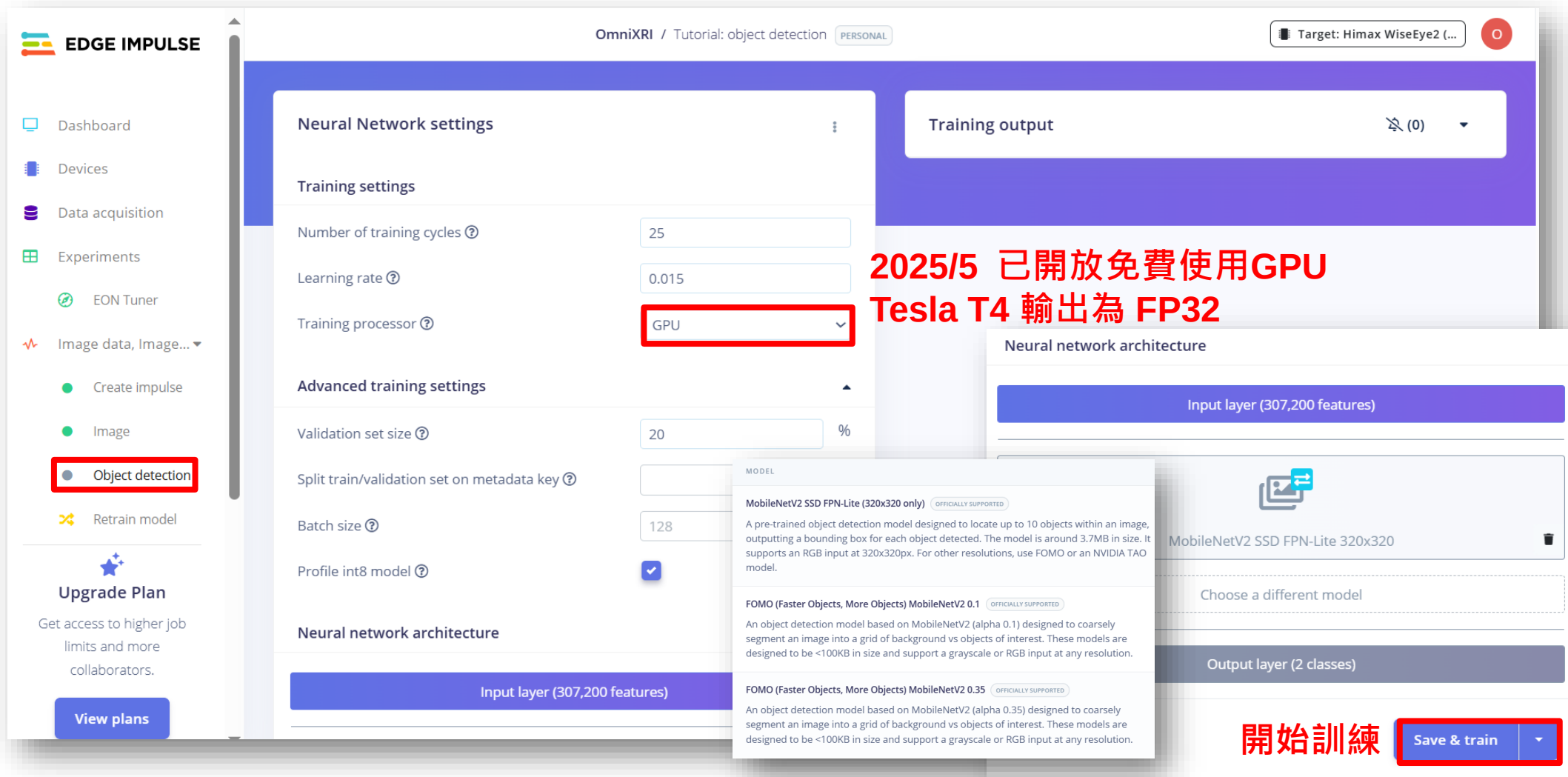
The screenshot displays the Edge Impulse Studio web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, and a dropdown menu for 'Image data, Image...'. Below these is an 'Upgrade Plan' section. The main area is titled 'OmniXRI / Tutorial: object detection' and 'PERSONAL'. It features a purple header bar with the text 'Image data, Image, Object Detection (Images)' and a description: 'An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.' Below this are four colored blocks: 'Image data' (red) with input axes 'image', 'Image width' (320), 'Image height' (320), and 'Resize mode' (Fit shortest a); 'Image' (white) with name 'Image' and input axes (1) 'image'; 'Object Detection (Images)' (purple) with name 'Object detection', input features 'Image', and output features '2 (coffee, lamp)'; and 'Output features' (green) with '2 (coffee, lamp)'. A 'Save Impulse' button is located at the bottom right of the main area.

Edge Impulse Studio — 產生特徵



The screenshot displays the Edge Impulse Studio web interface. On the left is a sidebar with navigation options: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Image data, Image..., Create impulse, Image (highlighted with a red box), Object detection, and Retrain model. Below these is an 'Upgrade Plan' section. The main area has a top bar with 'OmniXRI / Tutorial: object detection' and a 'PERSONAL' badge. A 'Target: Himax WiseEye2 (...)' dropdown is in the top right. The 'Parameters' tab is active, with 'Generate features' highlighted in a red box. Below this, the 'Training set' section shows 'Data in training set' as 31 items and 'Classes' as 2 (coffee, lamp). A 'Generate features' button is highlighted in a red box. The 'Feature explorer' on the right shows a scatter plot with blue dots for 'coffee' and orange dots for 'lamp'. The 'Feature generation output' section at the bottom shows a muted speaker icon and '(0)' items.

Edge Impulse Studio — 訓練模型



EDGE IMPULSE

OmniXRI / Tutorial: object detection PERSONAL

Target: Himax WiseEye2 (...)

Neural Network settings

Training settings

Number of training cycles 25

Learning rate 0.015

Training processor GPU

Advanced training settings

Validation set size 20 %

Split train/validation set on metadata key

Batch size 128

Profile int8 model

Neural network architecture

Input layer (307,200 features)

MobileNetV2 SSD FPN-Lite 320x320

Choose a different model

Output layer (2 classes)

MobileNetV2 SSD FPN-Lite (320x320 only) OFFICIALLY SUPPORTED

A pre-trained object detection model designed to locate up to 10 objects within an image, outputting a bounding box for each object detected. The model is around 3.7MB in size. It supports an RGB input at 320x320px. For other resolutions, use FOMO or an NVIDIA TAO model.

FOMO (Faster Objects, More Objects) MobileNetV2 0.1 OFFICIALLY SUPPORTED

An object detection model based on MobileNetV2 (alpha 0.1) designed to coarsely segment an image into a grid of background vs objects of interest. These models are designed to be <100KB in size and support a grayscale or RGB input at any resolution.

FOMO (Faster Objects, More Objects) MobileNetV2 0.35 OFFICIALLY SUPPORTED

An object detection model based on MobileNetV2 (alpha 0.35) designed to coarsely segment an image into a grid of background vs objects of interest. These models are designed to be <100KB in size and support a grayscale or RGB input at any resolution.

2025/5 已開放免費使用GPU
Tesla T4 輸出為 FP32

開始訓練

Save & train

Edge Impulse Studio — 訓練結果

Training output

(0)

Model training complete

Loading data for profiling...

Using batch size: 128

Loading data for profiling OK

Calculating performance metrics...

Calculating inferencing time...

INFO: Created TensorFlow Lite XNNPACK delegate for CPU.

Calculating inferencing time OK

Calculating float32 accuracy...

Job completed (success)

Model

Model version: ? Unoptimized (float32)

Last training performance (validation set)

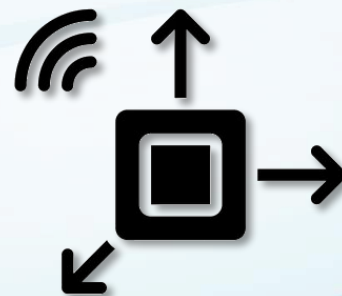
%

PRECISION SCORE ?

89.4%

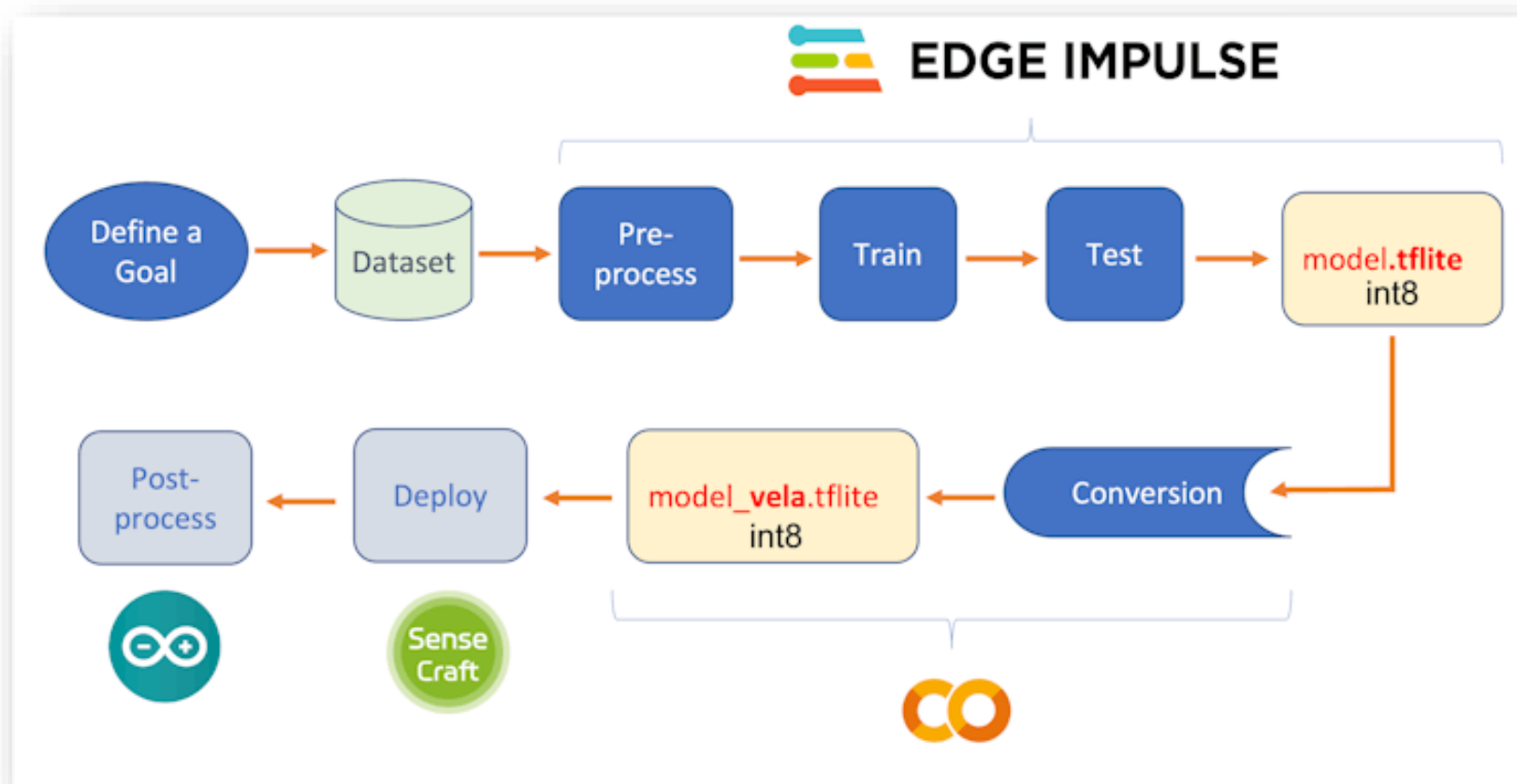
Metrics (validation set)

METRIC	VALUE
mAP ?	0.62
mAP@[IoU=50] ?	1.00
mAP@[IoU=75] ?	0.68
mAP@[area=small] ?	-1.00
mAP@[area=medium] ?	0.77
mAP@[area=large] ?	0.57
Recall@[max_detections=1] ?	0.70
Recall@[max_detections=10] ?	0.70
Recall@[max_detections=100] ?	0.70
Recall@[area=small] ?	-1.00
Recall@[area=medium] ?	0.80
Recall@[area=large] ?	0.64



11.4. 模型部署與測試

採用 Edge Impulse 訓練轉換



<https://www.hackster.io/mjrobot/computer-vision-at-the-edge-with-grove-vision-ai-module-v2-0003c7>

下載 TFLite 模型



The screenshot shows the Edge Impulse dashboard interface. On the left, the 'Dashboard' menu item is highlighted with a red box. The main content area displays 'About this project' and a 'Download block output' section. A table lists available downloads:

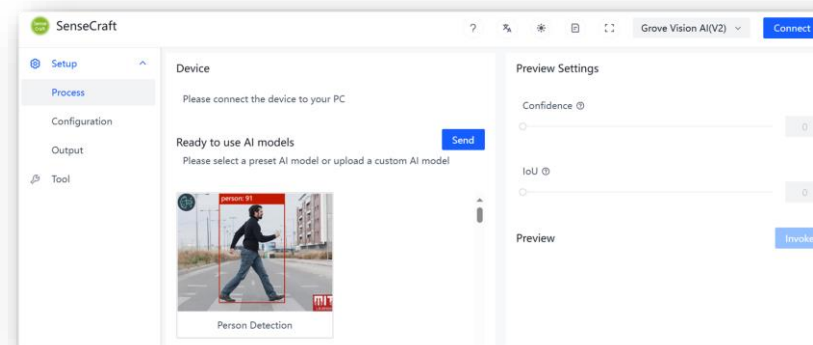
TITLE	TYPE	SIZE
Image training data	NPY file	31 windows
Object detection model	TensorFlow Lite (int8 quantized)	3 MB
Object detection model	TensorFlow SavedModel	10 MB

A red box highlights the 'Object detection model' row with TensorFlow Lite (int8 quantized) type and 3 MB size. A red arrow points to the download icon (a document with a download symbol) next to this row. A yellow text box is overlaid on the table, providing instructions on how to rename the downloaded file.

**下載原始名稱 ei-tutorial_-object-detection-object-detection-tensorflow-lite-int8-quantized-model.5.lite
更名為較短名稱並將附檔名變為 tflite
ei-object-detection-int8.tflite**

Below the table, a white box contains the text: **Object detection model** and **TensorFlow Lite (int8 quantized)**.

使用 Google Colab 進行轉換



```
!pip install ethos-u-vela
!!vela --version
```

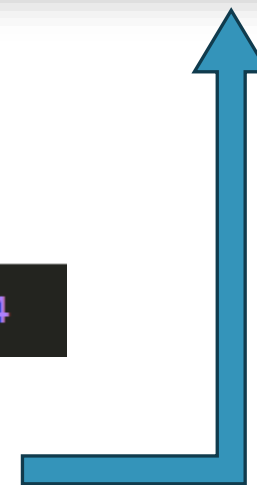
安裝 Arm Ethos-U NPU 編譯器

上傳 ei-object-detection-int8.tflite 到 Colab 檔案夾

```
!vela ei-object-detection-int8_vela.tflite --accelerator-config ethos-u55-64
```

開始編譯產生 ei-object-detection-int8_vela.tflite

使用 SenseCraft Web toolkit 將檔案燒錄到 Grove Vision AI V2
(參考第17頁)



參考文獻

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- 許哲豪，OmniXRI's Edge AI & TinyML 小學堂 Youtube 直播課程總結
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- 許哲豪，歐尼克斯實境互動工作室系列發文—TinyML(MCU AI)系列
<https://hackmd.io/1PK1URhIQ7GutcWgpgsWbg#TinyMLMCU-AI%E7%B3%BB%E5%88%97>
- Edge Impulse, Seeed Grove Vision AI Module V2 (WiseEye2)
<https://docs.edgeimpulse.com/docs/edge-ai-hardware/mcu-+-ai-accelerators/himax-seeed-grove-vision-ai-module-v2-wise-eye-2>
- Edge Impulse, Object detection with bounding boxes
<https://docs.edgeimpulse.com/docs/tutorials/end-to-end-tutorials/computer-vision/object-detection/object-detection>

延伸閱讀

- Edge Impulse CLI Installation

<https://docs.edgeimpulse.com/docs/tools/edge-impulse-cli/cli-installation>

- Seeed Studio Grove Vision AI Module V2

https://wiki.seeedstudio.com/grove_vision_ai_v2/

- Seeed SenseCraft Web Toolkit

<https://seeed-studio.github.io/SenseCraft-Web-Toolkit/#!/setup/process>

- Github – edgeimpulse, firmware-seeed-grove-vision-ai-module-v2

<https://github.com/edgeimpulse/firmware-seeed-grove-vision-ai-module-v2>

- MJRoBot (Marcelo Rovai), MJRoBot (Marcelo Rovai)Computer Vision at the Edge with Grove Vision AI Module V2

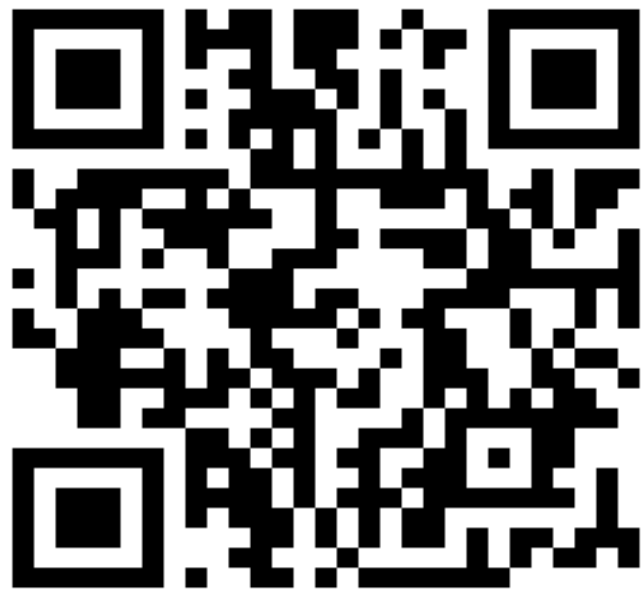
<https://www.hackster.io/mjrobot/computer-vision-at-the-edge-with-grove-vision-ai-module-v2-0003c7>

沒有最邊



只有更邊

歡迎加入
邊緣人俱樂部



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開 源 : <https://github.com/OmniXRI>

YOUTUBE 直播 : <https://www.youtube.com/@omnixri1784/streams>